

Main PCB

Power:

12V

- Power PCB 12V → Switch → Capacitor + → 12V Rail
- Power PCB GND → Capacitor - → GND rail

5V

- Power PCB 5V → Switch → Capacitor + → 5V Rail
- Power PCB GND → Capacitor - → GND rail

3.3V

- Power PCB 3.3V → Switch → Capacitor + → 3.3V Rail
- Power PCB GND → Capacitor - → GND rail

Teensy 4.1

- VIN → 5V Rail
- GND → GND rail

Sensors etc:

Gyro(Adafruit BNO055)

- VIN → 3.3V
- GND → GND rail
- SCL → Teensy **pin 24**
- SDA → Teensy **pin 25**

Ultrasonic(HC-SR04)

- VCC → 3.3V
- Trig → Teensy **pin 20**
- Echo → Teensy **pin 21**
- GND → GND rail

Relay Module

- VCC → 5V Rail
- IN1 → Teensy **pin 22**
- GND → GND_Rail

Line PCB

- TX_L → RX8
- RX_L → TX8

IR PCB

- TX_IR → RX4
- RX_IR → TX4

Oled Screen

- SDA_S → Teensy pin 18
- SCL_S → Teensy pin 19

Ultrasonic PCB

- TX_U → 15
- RX_U → 14

RoboCup Communication Module

- <https://github.com/robocup-junior/soccer-communication-module>
GND → GND rail
LOGV → 3.3V rail
- Start/Stop outputs (GO/STOP)
OUT1 → Teensy A12
OUT2 → Teensy A13
- UART (Serial7)
TX (module) → RX7 (Teensy)
RX (module) → TX7 (Teensy)

2) Motor drivers

DRV8263H #1 (Motor 1)

Power

- VM/VIN → **12V rail**
- GND → **GND rail**
- VDD → **3.3V rail**

Mode / enable (recommended)

- MODE → **GND** (*PH/EN style using IN1/IN2 pins*)
- SR → **GND**
- DRVOFF → **GND**
- SLEEP → **Teensy pin 6** (*shared with all drivers*)

Control

- 1_IN1 → **Teensy pin 2**
- 1_IN2 → **Teensy pin 3**

Motor

- OUT1, OUT2 → **Pololu 12V 1.6A Motor 1**

DRV8263H #2 (Motor 2)

Power

- VM/VIN → **12V rail**
- GND → **GND rail**
- VDD → **3.3V rail**

Mode / enable (recommended)

- **MODE → GND**
- **SR → GND**
- **DRVOFF → GND**
- **SLEEP → Teensy pin 6** (*shared*)

Control

- **2_IN1 → Teensy pin 4**
- **2_IN2 → Teensy pin 5**

Motor

- **OUT1, OUT2 → Pololu 12V 1.6A Motor 2**

DRV8263H #3 (Motor 3)

Power

- **VM/VIN → 12V rail**
- **GND → GND rail**
- **VDD → 3.3V rail**

Mode / enable (recommended)

- **MODE → GND**
- **SR → GND**
- **DRVOFF → GND**
- **SLEEP → Teensy pin 6** (*shared*)

Control

- 3_IN1 → **Teensy pin 9**
- 3_IN2 → **Teensy pin 10**

Motor

- OUT1, OUT2 → **Pololu 12V 1.6A Motor 3**

DRV8263H #4 (Motor 4)

Power

- VM/VIN → **12V rail**
- GND → **GND rail**
- VDD → **3.3V rail**

Mode / enable (recommended)

- MODE → **GND**
- SR → **GND**
- DRVOFF → **GND**
- SLEEP → **Teensy pin 6** (*shared*)

Control

- 4_IN1 → **Teensy pin 11**
- 4_IN2 → **Teensy pin 12**

Motor

- OUT1, OUT2 → **Pololu 12V 1.6A Motor 4**

Optional (leave unconnected if you don't need them)

- FAULT → Teensy input (optional)
- IPROPI → Teensy analog input (optional current sense)
- DIAG → optional / leave default